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# Accurate and Reliable Prediction of the Binding Affinities of Macrocycles to Their Protein Targets

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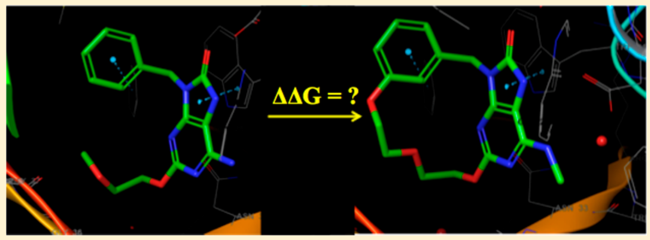
*Supporting Information*

**ABSTRACT:** Macrocycles have been emerging as a very important drug class in the past few decades largely due to their expanded chemical diversity benefiting from advances in synthetic methods. Macrocyclization has been recognized as an effective way to restrict the conformational space of acyclic small molecule inhibitors with the hope of improving potency, selectivity, and metabolic stability. Because of their relatively larger size as compared to typical small molecule drugs and the complexity of the structures, efficient sampling of the accessible macrocycle conformational space and accurate prediction of their binding affinities to their target protein receptors poses a great challenge of central importance in computational macrocycle drug design. In this article, we present a novel method for relative binding free energy calculations between macrocycles with different ring sizes and between the macrocycles and their corresponding acyclic counterparts. We have applied the method to seven pharmaceutically interesting data sets taken from recent drug discovery projects including 33 macrocyclic ligands covering a diverse chemical space. The predicted binding free energies are in good agreement with experimental data with an overall root-mean-square error (RMSE) of 0.94 kcal/mol. This is to our knowledge the first time where the free energy of the macrocyclization of linear molecules has been directly calculated with rigorous physics-based free energy calculation methods, and we anticipate the outstanding accuracy demonstrated here across a broad range of target classes may have significant implications for macrocycle drug discovery.




## 1. INTRODUCTION

Macrocycles are defined herein as molecules containing at least one large ring with more than 10 heavy atoms along the backbone of the ring. Such molecules occupy a unique region in chemical space, bridging the gap between druglike small molecules and larger biomolecules, such as biologically active peptides and proteins. The molecular weights of macrocycles are larger than typical druglike small molecules, so to are their surface areas and the numbers of hydrogen bond donors and acceptors. Macrocyclization restricts internal bond rotation and thus the available conformational space, leading to their unique ability to span a larger binding site surface area while remaining conformationally restricted as compared to their acyclic counterparts with similar molecular weights. These features make macrocycles a promising modality for targeting protein−protein interfaces and other shallow or poorly defined binding sites.[1] Further, macrocyclization is often an attractive strategy in pharmaceutical drug discovery because of the provided broad ability to modulate potency, selectivity, membrane permeability, and many other relevant physicochemical properties.[1,2]

There are four classes of macrocycles, including peptidic and nonpeptidic natural products, non-natural peptides, and non-natural macrocycles.[1,2] Upon macrocyclization, the potency of a certain ligand can either increase or decrease depending on the specific interactions it has with the protein and the conformations it adopts. On the one hand, the restricted conformational space of macrocycles, as compared to the linear molecules, may minimize the unfavorable reduction of entropy from the internal degrees of freedom experienced when binding to the receptors, leading to increased binding potency versus the acyclic counterpart molecule. On the other hand, macrocyclization may also result in less favorable interactions with protein due to increased rigidity of the ligands or increased static clashes in the bound complex as compared to the acyclic counterpart, or the bioactive conformation may be a high-energy state with large unfavorable internal strain energy for the macrocycle; these effects will reduce the binding potency upon macrocyclization. Although macrocycles are typically more challenging to synthesize than their corresponding acyclic counterparts, and macrocyclization leading to enhanced binding potency is rare through brute force enumeration, macrocyclization strategies are often attempted in drug discovery programs due to their unique ability to balance various important properties of the compounds including, potency, selectivity, metabolic stability, and bioavailability.[2] In certain cases, for example for protein−protein interfaces, macrocycles might be one of only a few viable choices for these challenging targets.[1] Therefore, accurate prediction of whether macrocyclization of an acyclic inhibitor might be beneficial, and

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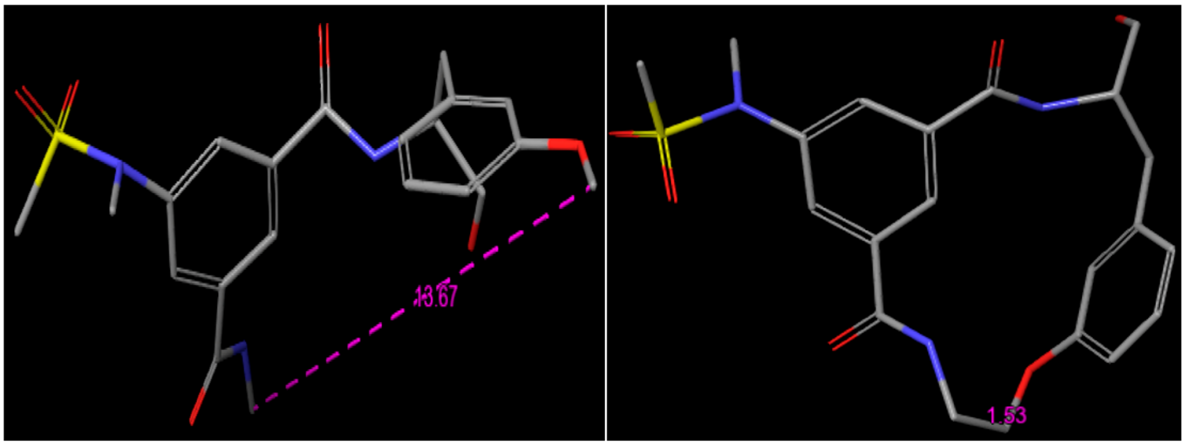



**Figure 1.** A pair of BACE-1 ligands with a macrocyclization transformation. (left) acyclic ligand 3; (right) macrocycle ligand 4. The macrocyclization is happening between the two carbon atoms. In the macrocycle, the distance between the two carbon atoms is ∼1.53A, whereas in acyclic ligand 3, the distance between these two carbon atoms is very long at ∼13.67 Å.

identifying the optimal size and chemical matter for macrocyclization, is of central importance in macrocycle drug design.

Previous efforts in macrocycle modeling have mainly been focused on the sampling of macrocycle conformational space.[3−9] However, in these prior studies, quantitative prediction of the binding potencies of the macrocycles, which is of central importance in computational macrocycle drug design, was not attempted. In the present paper, we present an extension to the FEP+ free energy calculation program,[10] which enables efficient sampling of macrocycle conformations and the accurate prediction of the binding affinities between the macrocycles and their protein targets. This method is based upon a previously established method, core-hopping FEP,[11] which was developed to model scaffold hopping transformations in small molecule drug discovery. In this approach, one of the bonds in a macrocycle is alchemically annihilated when transforming into an acyclic molecule or into a different macrocycle with a different ring size. In this way, a macrocycle with an appropriately alchemically softened bond in the intermediate lambda windows in macrocycle FEP simulations can efficiently sample different conformations, and through efficient Monte Carlo replica exchange, these different conformations are propagated to the physical end points with the correct Boltzmann weighting.[10,12−14]

In the following, we present the novel macrocycle FEP methodology, demonstrate the use of the method on seven different pharmaceutically interesting data sets where a diverse set of macrocyclization strategies have been attempted to achieve the target potency, and discuss the implications of the method for macrocycle drug design.

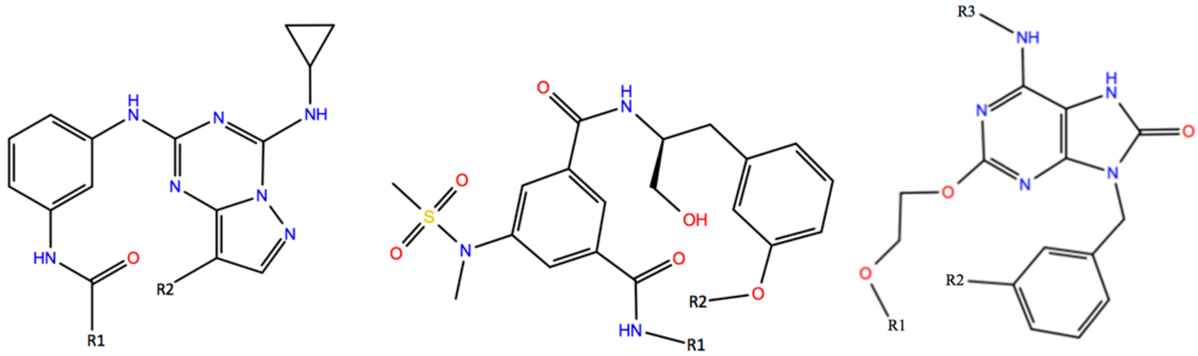
## 2. METHODS

Free energy perturbation (FEP) is a rigorous method for relative protein−ligand binding free energy calculations based on first principle statistical mechanics.[10,15−20] Traditional relative binding affinity FEP methods can only be used to calculate the relative binding free energies between molecules that differ by functional groups (R-group mutations) or between molecules that differ by single atom changes (atom mutations).[11−13] In the applications of FEP to macrocycle design, the most crucial initial problem is to determine if the initial linear molecule will maintain or improve its binding affinity should it be macrocyclized, and further, what the optimal length and chemical composition of the linker group are that should be used to macrocyclize the linear molecule. Both of these design goals require the calculation of free energy to form and/or break a covalent bond on the macrocycle backbone. This is not possible through the traditional relative binding affinity FEP methods.[11,21]

In 2016, we reported a novel core-hopping FEP method that uses a soft bond potential enabling the calculation of free energy to break or form a covalent bond through FEP simulations.[11,22] The core-hopping FEP method has been used successfully to predict binding free energy changes due to scaffold hopping modifications in small molecule drug discovery projects.[11,18] Here, we adopt the same soft bond potential functional form used in core-hopping FEP to calculate the free energy to form or break a covalent bond within macrocycles, enabling rigorous free energy calculations for macrocyclization perturbations and ring size optimization of macrocycles.

In small molecule scaffold hopping modifications, the sizes of the rings that change topologies in the transformations are relative small, typically only 4−8 heavy atoms. In such cases, although the equilibrium distance between two atoms connected by a soft bond will be longer when the bond is annihilated than when the bond is present, the distance change is still relatively small (most often only a few angstroms) due to the small size of the ring containing the atoms sharing the soft bond. In macrocycle perturbations, however, the equilibrium distance between the two atoms sharing the soft bond can be very long in the acyclic molecule due to the large size of the macrocyclic ring. This can be clearly seen in Figure 1 for a typical macrocyclization perturbation between a pair of BACE-1 ligands, where ligand 3 is acyclic and ligand 4 has been macrocyclized through introduction of a covalent bond between the methoxy group and the distal methyl-amide group.[23] In the acyclic species (ligand 3), the equilibrium distance between these two relevant carbon atoms (∼13.67 Å) is very long compared with the equilibrium distance (1.53 Å) observed for macrocyclic species (ligand 4). Therefore, to achieve good phase space overlap between the two end points in macrocyclization FEP simulations, which is essential for converged free energy calculations, the soft bond potential used for macrocycle FEP calculations must be adjusted so that both the long and short bond distances can be sampled efficiently in

Table 1. Experimental and Predicted Binding Potencies (kcal/mol) for Macrocyclization Transformations of CK2, BACE-1, and MTH1 Inhibitors




| compound | $R_1-R_2$ | exp. dG | pred. dG | CCC error[a] |
|---|---|---|---|---|
| | | CK2 | | |
| 1 | $R_1 = CH_3$, $R_2 = CN$ | −13.04 | −13.24 | 0.33 |
| 2 | $R_1-R_2 = -(CH_2)_4-$ | −10.36 | −10.15 | 0.33 |
| | | BACE-1 | | |
| compound | $R_1-R_2$ | exp. dG | pred. dG | CCC error[a] |
| 3 | $R_1,R_2 = CH_3$ | −5.44 | −5.62 | 0.31 |
| 4 | $R_1-R_2 = -(CH_2)_2-$ | −7.53 | −7.35 | 0.31 |
| | | MTH1 | | |
| compound | $R_1-R_3$ | exp. dG | pred. dG | CCC error[a] |
| 5 | $R_1,R_3 = CH_3$; $R_2 = H$ | −7.57 | −6.86 | 0.24 |
| 6 | $R_1 = CH_3$; $R_2,R_3 = H$ | −8.55 | −8.69 | 0.23 |
| 7 | $R_1-R_2 = -CH_2-O-$; $R_3 = CH_3$ | −12.69 | −13.26 | 0.24 |

[a]Cycle closure correction error, which is computed following the algorithm in refs 14 and 27.

the intermediate lambda windows during the FEP simulations. In the Supporting Information, we show that the soft bond parameter $\alpha$ in the soft bond potential introduced in the core-hopping FEP method controls the effective range of distances of the soft bond sampled in the intermediate lambda windows; a larger alpha results in a shorter range of sampled distances, and a smaller alpha results in a longer range of sampled distances. Although a soft bond parameter of 2 is used in core-hopping FEP because of the small change in the soft bond distances in small molecule scaffold hopping transformation, we find that a smaller soft bond parameter of 0.5 is needed for macrocycle FEP due to the large soft bond distance changes between the two molecules, which works well for all the transformations tested in this study. Detailed discussion about the soft bond potential, the soft bond parameters, and the connections between the macrocycle FEP method and the previously reported core-hopping FEP methodology are included in the Supporting Information.

**Details of the Simulation Protocols.** The macrocycle FEP method is implemented in the FEP+ program in Schrodinger Maestro Suite 2017[10,24] and used for all of the relative binding free energy calculations reported here. The OPLS3 force field[25] is used to describe the proteins and ligands, and the water molecules are modeled with SPC potential.[26] Cycle closure algorithms are used to assess the convergence of the simulations and estimate sampling error.[14,27] The initial structures used for the simulations are taken from PDB with PDB IDs 2E9P,[28] 2PVN,[29] 2B8V,[30] 2Q15,[31] 5ANT,[32] 3RKZ,[33] and 3VHA.[34] The proteins are prepared with Protein Preparation Wizard,[35] and the protonation states are assigned with the same pH reported in the experimental binding affinity assays. The noncrystal ligands are aligned to the corresponding crystal ligands by using the Flexile Ligand Assignment tool in Maestro. The systems are relaxed and equilibrated with the default protocol in FEP+.[10,11] All simulations were run with 16 lambda windows and 25 ns production simulation for each lambda window. Free energy results were obtained by using the entire 25 ns simulation data and the first 5 ns data to assess the convergence of the simulations.

The sampling error for each perturbation is calculated using the Bennett acceptance ratio method. For systems with more than three ligands, closed cycles are constructed in the perturbation pathways, and the cycle closure method is used to assess the sampling error. The cycle closure-corrected relative binding free energies among the ligands are converted to absolute binding free energies of the ligands following the same method reported in ref 10. In this conversion, the average of the predicted binding free energies is set to be equal to the average of the experimental binding free energies. This has no effect on correlation of the predicted values with the experimental data

## 3. RESULTS AND DISCUSSION

In this section, we present seven pharmaceutically interesting data sets in which diverse macrocyclization strategies have been attempted to optimize the lead compound. The macrocycle systems studied in this paper include one series of CHK1 inhibitors,[28] one series of CK2 inhibitors,[29] two series of BACE-1 inhibitors,[23,31,36] one series of MHT1 inhibitors,[32] and two series of Hsp90 inhibitors.[33,34] We then present the relative binding free energy calculation results for these systems using the macrocycle FEP method described in the Methods section



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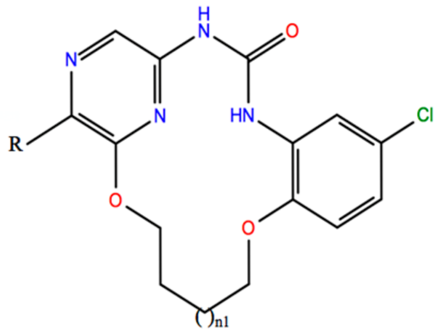
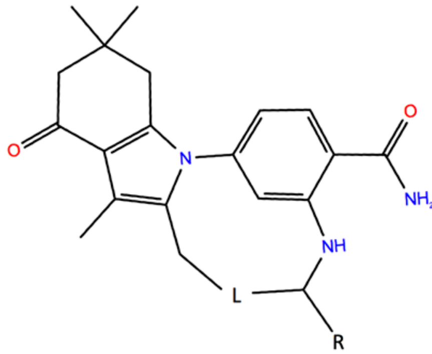
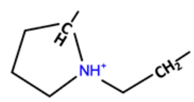
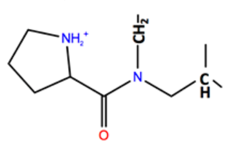
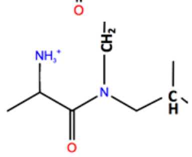
**Table 2. Experimental and Predicted Binding Potencies (kcal/mol) for Macrocycle Ring Expansion Transformations of CHK1 and Hsp90 Inhibitors**



CHK1 (ligand 8-11)



Hsp90 (ligand 12-18)

| Comp. | CHK1 | | | | |
|---|---|---|---|---|---|
| | R | n1 | Exp. dG | Pred. dG | CCC Error[a] |
| 8 | H | 2 | -10.88 | -10.00 | 0.21 |
| 9 | CN | 1 | -11.18 | -11.15 | 0.21 |
| 10 | CN | 2 | -11.09 | -11.21 | 0.21 |
| 11 | CN | 3 | -10.27 | -11.05 | 0.22 |
| | Hsp90 | | | | |
| | L | R | Exp. dG | Pred. dG | CCC Error[a] |
| 12 | $–CH_2–NH_3^+–CH_2–$ | H | -6.67 | -6.27 | 0.18 |
| 13 | $–C(CH_3)_2–NH_2^+–CH_2–CH_2–$ | H | -9.32 | -8.86 | 0.17 |
| 14 | $–C(CH_3)_2–CH_2–NH_2^+–CH_2–CH_2–$ | H | -8.89 | -8.78 | 0.18 |
| 15 | $–CH_2–NH^+(CH_3)–CH_2–CH_2–$ | $CH_3$ | -9.28 | -9.29 | 0.17 |
| 16 |  | H | -8.45 | -9.15 | 0.17 |
| 17 |  | $CH_3$ | -13.14 | -13.23 | 0.18 |
| 18 |  | $CH_3$ | -12.74 | -12.90 | 0.18 |

[a]Cycle closure correction error, which is computed following the algorithm in ref 14 and 27.

and compare the predictions with experimental data. Three-dimensional structures of individual ligands and the protein targets used as the input of the macrocycle FEP calculations and output files of all the calculations are included in the Supporting Information. Two-dimensional structures of these ligands are also shown in Tables 1−3 along with the calculated and experimental binding free energy results. These modifications cover two of the most important and common transformations of high interest in macrocycle drug design. These include (1) the transformation from an acyclic molecule to a macrocycle molecule, which is very important when assessing whether a macrocyclization strategy is likely to succeed to maintain or improve potency in a drug discovery project, and (2) changing the size of the linker used to macrocyclize the acyclic molecule, which is important to identify the optimal length and chemical composition of the cyclizing linker group.

**3.1. Macrocyclization Transformations.** A macrocyclization perturbation here refers to a typical type of chemical transformation in macrocycle drug design projects where an acyclic molecule is transformed into a macrocycle with the introduction of an additional bond or linking moiety, where it is generally hoped that the restriction of torsional angles due to cyclization will reduce the entropy cost associated with the protein−ligand binding process, thus improving the binding potency and possibly the binding selectivity. Three sets of


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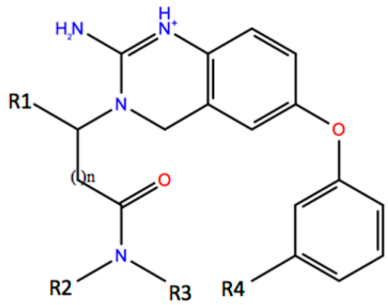
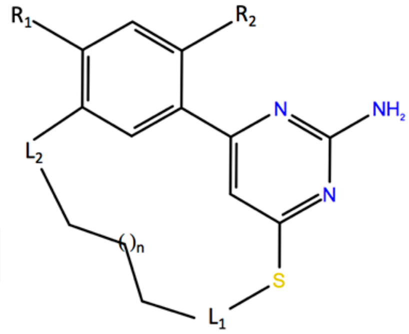
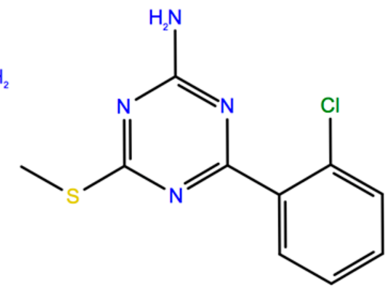
**Table 3. Transformations among Acyclic and Macrocycle Molecules Binding to BACE-1 and Hsp90 Receptors along with Macrocycle FEP Predicted Binding Free Energies and Experimental Data**



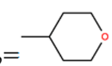

BACE-1 (Ligand 19-23)




Hsp90 (Ligand 24-32)




Hsp90 (Ligand 33)

| Comp. | BACE-1 | | | | |
|---|---|---|---|---|---|
| | $R_1$-$R_4$ | n1 | Exp. dG | Pred. dG | CCC Error[a] |
| 19 | $R_1$, $R_2$ = Cyclohexyl; $R_3$=$CH_3$; $R_4$=H | 1 | -10.82 | -10.33 | 0.48 |
| 20 | $R_1$=H, $R_2$= Cyclohexyl; –$R_3$–$R_4$– = –$(CH_2)_3$– | 1 | -9.82 | -10.22 | 0.42 |
| 21 | $R_1$, $R_2$ = Cyclohexyl; –$R_3$–$R_4$– = –$(CH_2)_2$– | 2 | -11.29 | -11.70 | 0.51 |
| 22 | $R_1$, $R_2$ = Cyclohexyl; –$R_3$–$R_4$– = –$(CH_2)_3$– | 2 | -10.21 | -9.86 | 0.41 |
| 23 | $R_1$= Cyclohexyl, $R_2$=  ; –$R_3$–$R_4$– = –$(CH_2)_2$– | 2 | -10.57 | -10.61 | 0.46 |

| | Hsp90 | | | | | |
|---|---|---|---|---|---|---|
| | L1-L3 | $R_1$-$R_2$ | n1 | Exp. dG | Pred. dG | CCC Error[a] |
| 24 | $L_1$= –1,3–$C_6H_4$–O–, $L_2$= –O– | $R_1$=H, $R_2$=$CH_3$ | 1 | -9.50 | -11.20 | 0.29 |
| 25 | $L_1$= –1,3–$C_6H_4$–O–, $L_2$= –O– | $R_1$=H, $R_2$=$CH_3$ | 2 | -10.15 | -11.48 | 0.29 |
| 26 | $L_1$= –1,3–$C_6H_4$–O–, $L_2$= –O– | $R_1$=H, $R_2$=$CH_3$ | 3 | -9.71 | -11.35 | 0.28 |
| 27 | $L_1$= –1,3–$C_6H_4$–O–, $L_2$= –NHCO– | $R_1$, $R_2$=$CH_3$ | 1 | -12.20 | -13.27 | 0.27 |
| 28 | $L_1$= –1,3–$C_6H_4$–O–, $L_2$= –NHCO– | $R_1$, $R_2$=$CH_3$ | 2 | -11.16 | -12.49 | 0.28 |
| 29 | $L_1$= –1,3–$C_6H_4$–O–, $L_2$= –NHCO– | $R_1$, $R_2$=$CH_3$ | 3 | -11.27 | -12.21 | 0.30 |
| 30 | $L_1$= –$(CH_2)_2$–NHCO–, $L_2$= –NHCO– | $R_1$, $R_2$=$CH_3$ | 1 | -12.74 | -11.30 | 0.28 |
| 31 | $L_1$= –$(CH_2)_2$–NHCO–, $L_2$= –NHCO– | $R_1$, $R_2$=$CH_3$ | 2 | -12.93 | -9.67 | 0.29 |
| 32 | $L_1$= –$(CH_2)_2$–NHCO–, $L_2$= –NHCO– | $R_1$, $R_2$=$CH_3$ | 3 | -13.18 | -11.32 | 0.31 |
| 33 | | | | -7.04 | -5.59 | 0.28 |

[a]Cycle closure correction error, which is computed following the algorithm in refs 14 and 27.

ligands involving macrocyclization transformations binding to CK2, BACE-1, and MTH1 receptors are shown in Table 1.

All three proteins studied here are important pharmaceutical drug targets. CK2 has been shown to play an important role in cell life cycle control, including the maintenance of cell viability and initiation of apoptosis, and inhibition of CK2 has been studied as a potential approach for cancer therapy.[37] The two CK2 inhibitors shown in Table 1 are taken from recent drug discovery programs.[38] BACE-1 plays a very important role in the formation of myelin sheaths in peripheral nerve cells. Recent clinical trials have targeted BACE-1 inhibitors as a potential therapeutic modality for the treatment of Alzheimer's disease.[39] Hydrogen bonding interactions between the BACE-1 inhibitors and the residues in the active site were shown to be important to tune the activities of the inhibitors, and the macrocyclization transformation shown in Table 1 has been shown to be an effective strategy to improve the binding potency.[23,40] MTH1 is one of the "housekeeping" enzymes that are responsible for hydrolyzing damaged nucleotides in cells and thus preventing them from being incorporated into DNA.[32] The MTH1 ligands shown in Table 1 are taken from a recent drug discovery project to modulate the activity of MTH1.[32]

The experimental and macrocycle FEP calculated binding free energies of these ligands and cycle closure correction errors are shown in Table 1. For BACE-1 and MTH1 inhibitors, macrocyclization of the acyclic molecules into macrocycles



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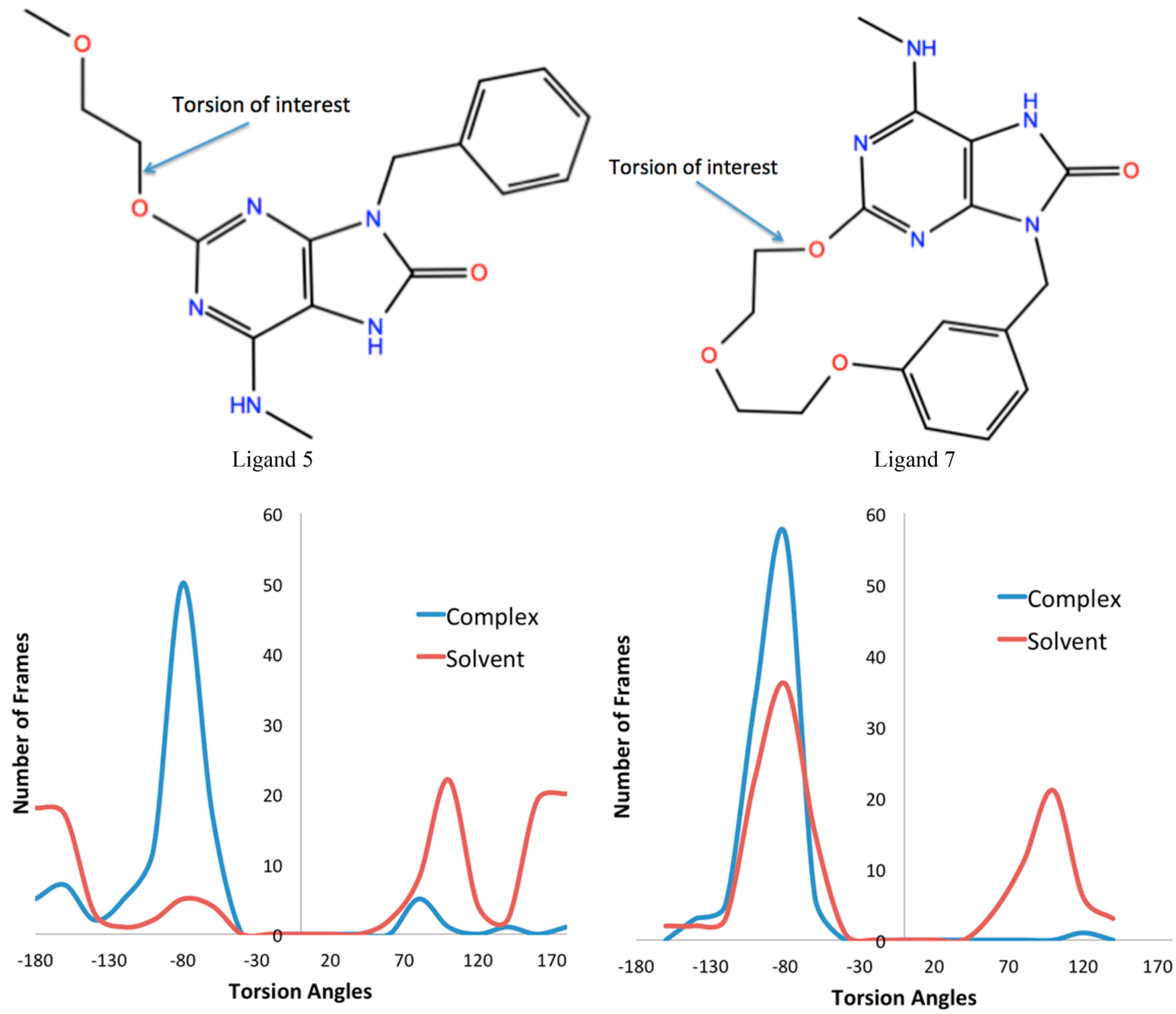


**Figure 2.** Torsional distributions of a key rotatable bond (labeled by an arrow) sampled in the complex and solvent legs of the macrocycle FEP simulations for the macrocyclization transformation from acyclic ligand 5 (left panel) to macrocycle ligand 7 (right panel) binding to the MHT1 receptor. A significant reduction of conformational entropy of the ligand is observed for the macrocyclization transformation, leading to the improved binding potency for the macrocycle as compared to the acyclic counterpart.

improves the binding potency dramatically in both cases, which is accurately modeled by macrocycle FEP calculations. In the case of BACE-1, addition of an ethyl linker to acyclic ligand 3 boosts the affinity of macrocyclic ligand 4 to be 34-fold more potent than the original species. Macrocycle FEP is able to accurately model this change in potency due to macrocyclization (calculated $\Delta\Delta G$ of −1.73 kcal/mol versus experimental $\Delta\Delta G$ of −2.09 kcal/mol). In the case of MTH1, macrocyclization of the acyclic molecule leads to 1000-fold improvement in potency, which is also correctly predicted by macrocycle FEP (predicted $\Delta\Delta G$ of −4.14 kcal/mol versus experimental $\Delta\Delta G$ of −4.57 kcal/mol). By comparison, for CK2, macrocyclization of acyclic molecule 14 into macrocycle 15 reduces the binding potency by 100-fold, suggesting a suboptimal macrocyclization design strategy was pursued in that work. This substantial loss of binding potency upon macrocyclization is also accurately predicted by macrocycle FEP (predicted $\Delta\Delta G$ of 2.68 kcal/mol versus experimental $\Delta\Delta G$ of 3.09 kcal/mol).

These examples demonstrate that macrocyclization can substantially increase or decrease the ligand binding potency depending on the specific macrocyclization strategy and the protein binding pocket. Several factors contribute to the binding free energy change upon macrocyclization. First, cyclization reduces the accessible conformational space of the molecules, and thus, the entropic cost upon binding to the protein pocket might be reduced, leading to an increase of binding potency. Second, macrocyclization may change the conformational free energy landscape of the molecule, making the bioactive conformation more or less stable, thus increasing or decreasing the binding potency. Third, macrocyclization may introduce static clash with the protein pocket decreasing the binding potency. These counterbalancing effects may result in an increase or decrease of the binding potency upon macrocyclization for any particular case.

The macrocycle FEP method takes into account all these aforementioned effects allowing a rigorous and accurate estimate of the binding free energy change upon macrocyclization. For the BACE-1 and MHT1 cases, the macrocycle FEP simulations indicate that macrocyclization of these molecules reduces their conformational entropy in bulk solution, leading to the improved potency for the macrocycles


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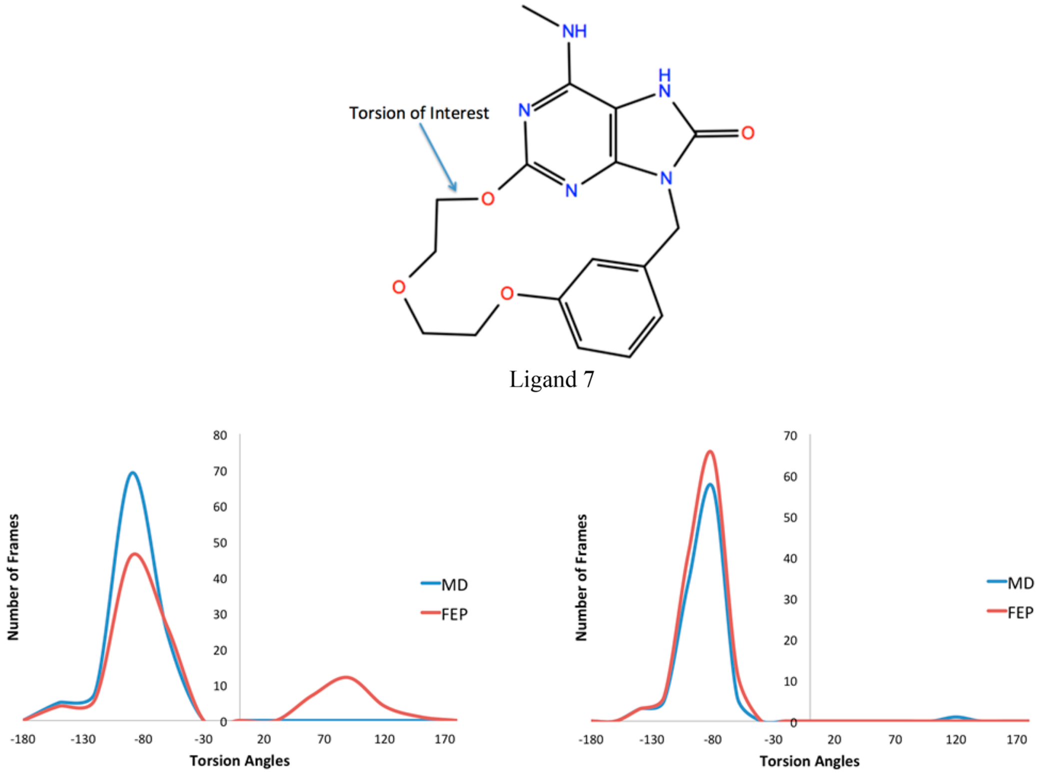


**Figure 3.** Distributions of a representative torsion (labeled by an arrow) of the MTH1 macrocycle inhibitor ligand 7 sampled in solvent (left panel) and complex (right panel) legs of the macrocycle FEP calculations. The corresponding torsion distributions sampled through brute force molecular dynamics (MD) simulations with the same account of simulation time (25 ns) are also shown for comparison. The red curves are from FEP simulations, and the blue curves are from MD simulations. Although similar ensembles of conformations are sampled for the complex phase, the macrocycle FEP samples many more conformations in the solvent phase than does brute force MD.

as compared to their acyclic counterparts. As an example, the torsional distributions of a key rotatable bond sampled in the complex and solvent legs of the macrocycle FEP simulations for the macrocyclization transformation from acyclic ligand 5 to macrocycle ligand 7 binding to the MHT1 receptor are shown in Figure 2. For acyclic ligand 5, a broad distribution of configurations is sampled in the solvent leg with three dominant conformations corresponding to torsion angles of approximately −80°, 100°, and 180°. However, in the protein−ligand complex, the interactions with the protein residues restrict the available conformations of the ligand, and only a very narrow distribution of configurations corresponding to a torsion angle of approximately −80° is sampled, resulting in a large loss of entropy during the protein−ligand binding process. In comparison, for macrocycle ligand 7, although the distribution of the relevant torsion angle sampled is similar in the protein−ligand complex (ie, one single dominant binding mode corresponding to a torsional angle of approximately −80°), the macrocyclization limits the available conformations in the unbound state as compared to the acyclic counterpart, and only two dominant configurations with torsion angles of approximately −80° and 100° are sampled for macrocycle ligand 7 in solution. Therefore, macrocyclization from ligand 5 to 7 significantly reduces the amount of entropy reduction during the protein−ligand binding process, leading to the improved binding potency. Quantitative characterization of the entropic contribution to the binding free energy can be obtained by multiple free energy simulations at different temperatures through the Van't Hoff equation or Shannon's conformational entropy formulation, both of which require significantly more simulations and will be the subject of future studies. Similarly, favorable entropic effects are also observed for the BACE-1 macrocyclization transformation. By comparison, for CK2 ligands, the specific macrocyclization strategy investigated here introduces steric clash between the linker moiety and the protein residues ILE66 and PHE 113, resulting in a substantial reduction in binding potency.

It is important to note that macrocyclization can introduce significant energy barriers to achieving a full sampling of the torsional space of macrocyclic ligands. In the macrocycle FEP method introduced here, efficient sampling of the macrocycle conformational space is achieved through the use of the soft bond potential coupled with lambda hopping replica exchange.[12,13] During the transformation of a macrocycle into an acyclic molecule, a covalent bond in the macrocycle is slowly annihilated through the soft bond potential resulting in the energy barriers for the rotation of the torsional angles being reduced in the intermediate lambda windows due to the partially broken soft bond, which in turn enables efficient sampling of torsional space of the macrocycle in these intermediate states. Through replica exchange, the different conformations sampled in the intermediate lambda windows are propagated into the physical end states, leading to the efficient sampling of the full conformational phase space of the macrocycles.

We presented an illustrative example in Figure 3 the distributions of a representative torsion of MTH1 macrocyclic ligand 7 sampled in solvent and complex legs of the macrocycle FEP calculations with the enhanced sampling scheme presented here. The corresponding torsional distributions sampled



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through brute force molecular dynamics (MD) simulations with the same account of simulation time (the sampling obtained if conventional FEP without lambda hopping were used) are also shown for comparison. It is clearly seen that the macrocycle FEP is able to sample many more conformations than MD, particularly in the solvent phases. In particular, the macrocycle FEP is able to sample a broad range of torsion angles with many transitions among the two dominant poses corresponding to torsion angles of approximately −80° and 100°; however, during the MD simulations, the ligand is trapped in the initial conformation corresponding to a torsion angle of ∼80° and does not sample the conformation with a torsion angle of approximately 100°. This clearly demonstrates the superior sampling efficiency of the macrocycle FEP method. In the complex, the interactions between the macrocycles and proteins limit the thermally accessible conformations of macrocycles, and the macrocycle FEP and MD sample roughly the same ensemble of conformations corresponding to the bioactive conformation of the macrocycle.

**3.2. Macrocycle Ring Size Change Transformations.** Macrocycle ring size change perturbations are also important transformations to consider in macrocycle drug discovery. These types of transformations involve the reduction or expansion of the size of the central macrocyclic ring to optimize the binding potency and selectivity or to balance other physicochemical properties of the molecules. Two sets of macrocycle ligands involving ring size change transformations binding to pharmaceutically important targets CHK1 and Hsp90 are shown in Table 2. CHK1 is involved in the coordination of the DNA damage response and cell cycle checkpoint response.[41] Ongoing clinical trials for CHK1 inhibitors mainly focus on their combination with gemcitabine,[28,41] which is a commonly prescribed chemotherapy. Hsp90 is also an important oncology target.[41,42] Both the CHK1 and Hsp90 ligands in Table 2 are taken from recent drug discovery programs pursuing these important drug targets.[31,34]

The experimental and macrocycle FEP calculated binding free energies and cycle closure errors of these ligands are shown in Table 2. For the CHK1 inhibitors, the binding potency is neither very sensitive to the size of the macrocycles (from 14- to 16-member rings) nor sensitive to minor chemical modifications on the linker (adding a Cl or CN group on the pyrazine ring). The macrocycle FEP correctly predicts that these molecules are roughly equipotent with an overall root-mean-square error (RMSE) of 0.76 kcal/mol for the five perturbations among ligands 8−11 as compared to experimental data.

In contrast, for the HSP90 molecules, the binding potency is very sensitive to the size of the macrocycles and to chemical modification of the linker moiety with the range of binding potencies of the macrocycles spanning from 12.44 uM to 290 nM. The macrocycle FEP calculations also correctly recapitulate the relative binding free energy differences among these molecules with an overall root-mean-square error (RMSE) of 0.51 kcal/mol for the nine perturbations among the seven ligands when compared to experimental data. The macrocycle FEP calculations also perfectly rank order these macrocycles and accurately recapitulate ligand 69 to be the most potent compound among these molecules.

We have also verified that the macrocycle conformational space is efficiently sampled for the ring size changes studied here. In the Supporting Information, we present the ensemble of structures of one of the HSP90 inhibitors, ligand 69, in solvent and complex phases, sampled by the macrocycle FEP and molecular dynamics (MD) with the same amount of simulation time. It is clearly observed that macrocycle FEP was able to sample many more conformations using the enhanced sampling scheme presented herein than were sampled when using simple MD for an equivalent simulation length for the same reasons discussed in the preceding section.

**3.3. Mixture of All Types for Transformations for Macrocycles.** In Table 3, we show two more series of ligands binding to BACE-1 and HSP90, respectively, where both macrocyclization and ring size transformations are required to model the series.[33,36] The BACE-1 inhibitors are taken from a recent publication by Huang et al. where the authors found a hairpin-like bioactive conformation for the linear molecule and suggested that formation of a macrocycle would improve the binding affinity through conformational restriction.[36] However, it was observed that some macrocyclization strategies when applied to acyclic molecule 19 improved the binding potency, whereas others diminished the binding potency.[36] Ligand 21 was the most potent macrocycle reported in this series of ligands with a seemingly optimal ring size and lipophilic cyclohexyl linker group. The macrocycle FEP calculations recapitulate this structure−activity relationship (SAR) and correctly indicate ligand 21 to be the most potent molecule. The overall RMSE of calculated relative binding free energies of these five ligands (ligands 19−23) is 0.54 kcal/mol. It is interesting to note that perturbations among ligands 21−23 have much large convergence errors than others (indicated by large hysteresis and large cycle closure errors for these perturbations). This is because the so-called flap loop in the BACE-1 receptor with which these ligands are interacting is known to be highly flexible and requires much longer sampling times to converge.

For the HSP90 inhibitors, all the reported macrocyclization strategies when applied to acyclic ligand 33 led to improved binding potency, which was also correctly recapitulated by the macrocycle FEP calculations. In addition, for each of the different macrocyclization strategies reported, macrocycle FEP was able to accurately rank order the relative binding free energies of the resulting macrocycles, including those involving the same chemistry but different linker lengths. For example, among macrocycles 24−26, which all have the same two ester groups on both sides of the linker, the macrocycle FEP is able to correctly predict ligand 25 to be the most potent binder. Similarly, among ligands 27−29, which all have an ester group on one side of the linker and an amide group on the other side of the linker, the macrocycle FEP also correctly identifies ligand 27 to be the most potent binder. However, for the transformations between different chemical groups on the linker, the macrocycle FEP predicted binding free energies have relatively large error as compared with experimental data, with the two perturbations, from ligand 27 to 30 and from ligand 32 to 29, having an error of ∼2.5 kcal/mol, both of which involve changing the ester group on the linker to an amide group. This could be due to the complex electrostatic interactions between the amide group on the linker and the polar residues in the protein binding pocket, which may not be precisely captured by the fixed charge force field employed. The overall RMSE of predicted relative binding affinities of these ten ligands is 1.35 kcal/mol.

**3.4. Summary of Macrocycle FEP Results and Comparison to a Simple Molecular Mechanics Ap-**


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**proach.** The macrocycle FEP calculated binding free energies of all 33 macrocycle ligands studied in the paper are reported in Table 4 with the distribution of the errors of the 38 perturbations plotted in Figure 4. Most of the perturbations have errors of less than 1 kcal/mol, and only 4 of out 38 have error larger than 1.5 kcal/mol, demonstrating the high accuracy of the reported method.

**Table 4. Experimental and Predicted Binding Potencies (kcal/mol) for 33 Ligands**

| ligand | exp. dG | pred. dG (5 ns) | pred. dG (25 ns) |
|---|---|---|---|
| 1 | −13.04 | −13.11 | −13.24 |
| 2 | −10.36 | −10.29 | −10.15 |
| 3 | −5.44 | −5.42 | −5.62 |
| 4 | −7.53 | −7.55 | −7.35 |
| 5 | −7.57 | −6.65 | −6.86 |
| 6 | −8.55 | −8.97 | −8.69 |
| 7 | −12.69 | −13.19 | −13.26 |
| 8 | −10.88 | −9.85 | −10.00 |
| 9 | −11.18 | −11.16 | −11.15 |
| 10 | −11.09 | −11.27 | −11.21 |
| 11 | −10.27 | −11.15 | −11.05 |
| 12 | −6.67 | −5.21 | −6.27 |
| 13 | −9.32 | −8.86 | −8.86 |
| 14 | −8.89 | −8.96 | −8.78 |
| 15 | −9.28 | −9.84 | −9.29 |
| 16 | −8.45 | −9.67 | −9.15 |
| 17 | −13.14 | −13.22 | −13.23 |
| 18 | −12.74 | −12.73 | −12.90 |
| 19 | −10.82 | −9.93 | −10.33 |
| 20 | −9.82 | −9.55 | −10.22 |
| 21 | −11.29 | −12.29 | −11.7 |
| 22 | −10.21 | −10.57 | −9.86 |
| 23 | −10.57 | −10.37 | −10.61 |
| 24 | −9.50 | −11.71 | −11.20 |
| 25 | −10.15 | −11.79 | −11.48 |
| 26 | −9.71 | −11.43 | −11.35 |
| 27 | −12.20 | −13.72 | −13.27 |
| 28 | −11.16 | −12.18 | −12.49 |
| 29 | −11.27 | −11.75 | −12.21 |
| 30 | −12.74 | −11.32 | −11.30 |
| 31 | −12.93 | −9.46 | −9.67 |
| 32 | −13.18 | −11.24 | −11.32 |
| 33 | −7.04 | −5.28 | −5.59 |

In Table 5, we also compared the macrocycle FEP results from the entire 25 ns simulations versus those from the first 5 ns simulations. Although the 25 ns results are slightly better than the 5 ns results, the difference is very small. The weighted MUEs for the 25 and 5 ns simulation results are 0.68 and 0.89 kcal/mol, respectively. The weighted RMSEs for the 25 and 5 ns simulation results are 0.94 and 1.11 kcal/mol, respectively. These results indicate that accurate predictions were obtained with 5 ns sampling and that the accuracy can be further improved with longer simulations. The relative binding free energies for all the perturbations studied in this paper are reported in Table S1.

The correlation plot between the macrocycle FEP predicted binding free energies and the experimental data for all of the 33 macrocycle ligands of the seven systems studied in this paper is shown in Figure S4 (left panel). These perturbations cover a very diverse set of macrocycles including both macrocyclization

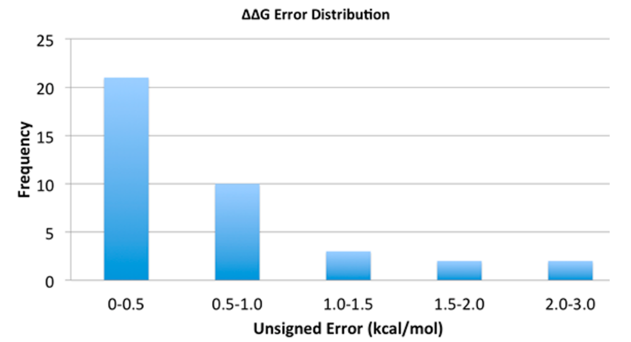


**Figure 4.** Error distribution of macrocycle FEP calculated binding free energies for 38 perturbations. Most of the predictions have errors less than 1 kcal/mol, and only 4 out of 38 perturbations have errors greater than 1.5 kcal/mol, demonstrating the highly accurate predictions of the method.

transformations and ring size change perturbations. The MUE and RMSE of the macrocycle FEP predicted relative binding free energies (ie, the computed ΔΔ$G$ values) as compared to experimental data are summarized in Table 5. The macrocycle FEP predicted binding free energies agree very well with the experimental data with an overall MUE and RMSE of 0.68 and 0.94 kcal/mol, respectively.

MM-GB/SA scoring results[43] for these ligands are also shown in Figure S4 (right panel) for comparison. Whereas the macrocycle FEP predicted binding free energies have high correlation with experimental data with a weighted average $R^2$ value of 0.78 for the three series of ligands with more than four data points, MM-GB/SA failed to capture many of the structure−activity relationships with a weighted average $R^2$ value of −0.07 for the same data set (MM-GB/SA scoring results are anticorrelated with experimental data for the second Hsp90 series of ligands). This is not surprising as the MM-GB/SA method cannot accurately capture the entropic effect in protein−ligand binding, which is one of the key components for the binding free energies of macrocycles.

## 4. CONCLUSIONS

Because of their unique ability to balance various important physicochemical properties including, potency, selectivity, membrane permeability, and bioavailability, macrocycles have been emerging as an important drug class with great potential to target shallow binding pockets, protein−protein interfaces, and other targets that are very difficult to drug using conventional small molecule therapeutics. One critically important challenge in macrocycle drug design is to assess whether a particular macrocyclization strategy is likely to succeed in improving, or at least maintaining, potency and to identify the optimal cyclizing linker moiety. The high cost and long time associated with the synthesis of macrocycles makes the accurate computational prediction of the binding free energies between the macrocycles and their target protein receptors crucially important for macrocycle drug discovery.

In this article, we present a novel macrocycle FEP method that is able to efficiently sample the accessible conformational space of macrocycles and accurately predict the relative binding affinities for various macrocycles of different central ring sizes as well as accurately assess the expected change in the binding free energy of an acyclic molecule should it be cyclized. We have validated the method on seven macrocyclic congeneric series involving diverse macrocycle transformations. The overall


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**Table 5. Summary of Error Statistics for the Macrocycle FEP Predicted Relative Binding Free Energies As Compared to Experimental Data**

| system | num. ligands | num. perturbations | PDB ID | $R^2$ (MMGBSA) | $R^2$ (FEP) 25 ns/5 ns | FEP (MUE) 25 ns/5 ns | FEP (RMSE) 25 ns/5 ns |
|---|---|---|---|---|---|---|---|
| CHK1 | 4 | 5 | 2E9P | | | 0.70/0.80 | 0.76/0.88 |
| CK2 | 2 | 1 | 2PVN | | | 0.41/0.15 | 0.41/0.15 |
| BACE-1 | 2 | 1 | 2B8V | | | 0.37/0.04 | 0.37/0.04 |
| BACE-1 | 5 | 6 | 2Q15 | 0.20 | 0.65/0.60 | 0.49/0.83 | 0.54/0.87 |
| MTH1 | 3 | 3 | 5ANT | | | 0.86/0.95 | 0.92/1.13 |
| Hsp90 | 7 | 9 | 3RKZ | 0.30 | 0.98/0.91 | 0.37/0.84 | 0.51/0.97 |
| Hsp90 | 10 | 13 | 3VHA | −0.46 | 0.37/0.32 | 0.97/1.08 | 1.35/1.41 |
| weighted average correlation ($R^2$) | | | | −0.07 | 0.78/0.73 | | |
| weighted average MUE ($\Delta\Delta G$) | | | | | | | 0.68/0.89 |
| weighted average RMSE ($\Delta\Delta G$) | | | | | | | 0.94/1.11 |

results of the macrocycle FEP calculated relative binding affinities agree very well with the experimental data with an MUE of 0.68 kcal/mol and an RMSE of 0.94 kcal/mol. For 35 out of the 38 perturbations studied here, the predicted binding free energies are within 1 kcal/mol of the experimental data. These results demonstrate that the macrocycle FEP method is able to accurately and reliably assess whether a macrocyclization strategy will improve the binding potency as well as help to identify the optimal cyclizing linker moiety.

We anticipate use of the macrocycle FEP method introduced here will have a significant impact on macrocycle drug discovery. Accurate prediction of the free energy changes for these common macrocycle transformations would not only save the synthetic effort wasted on the macrocycles that are unlikely to meet the targeted binding potency but would also encourage the chemist to pursue synthetic directions that would be considered as too challenging or risky without the benefit of the computational guidance. When considering the synthesis of an attractive but synthetically challenging macrocycle in a drug discovery project, if macrocycle FEP predicts the molecule to have a desired binding potency with the target protein, the risk of undertaking the synthetic challenge is substantially reduced, leading to more diverse chemical space being explored by the project team and rapid completion of difficult projects.

## ■ ASSOCIATED CONTENT

### Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jctc.7b00885.

Detailed discussion regarding the soft bond alpha parameter, comparison of macrocycle FEP results with MMGBSA results, ensembles of structures sampled in macrocycle FEP simulations, and the input and output files for macrocycle FEP simulations (PDF)

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### Notes

The authors declare no competing financial interest.

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